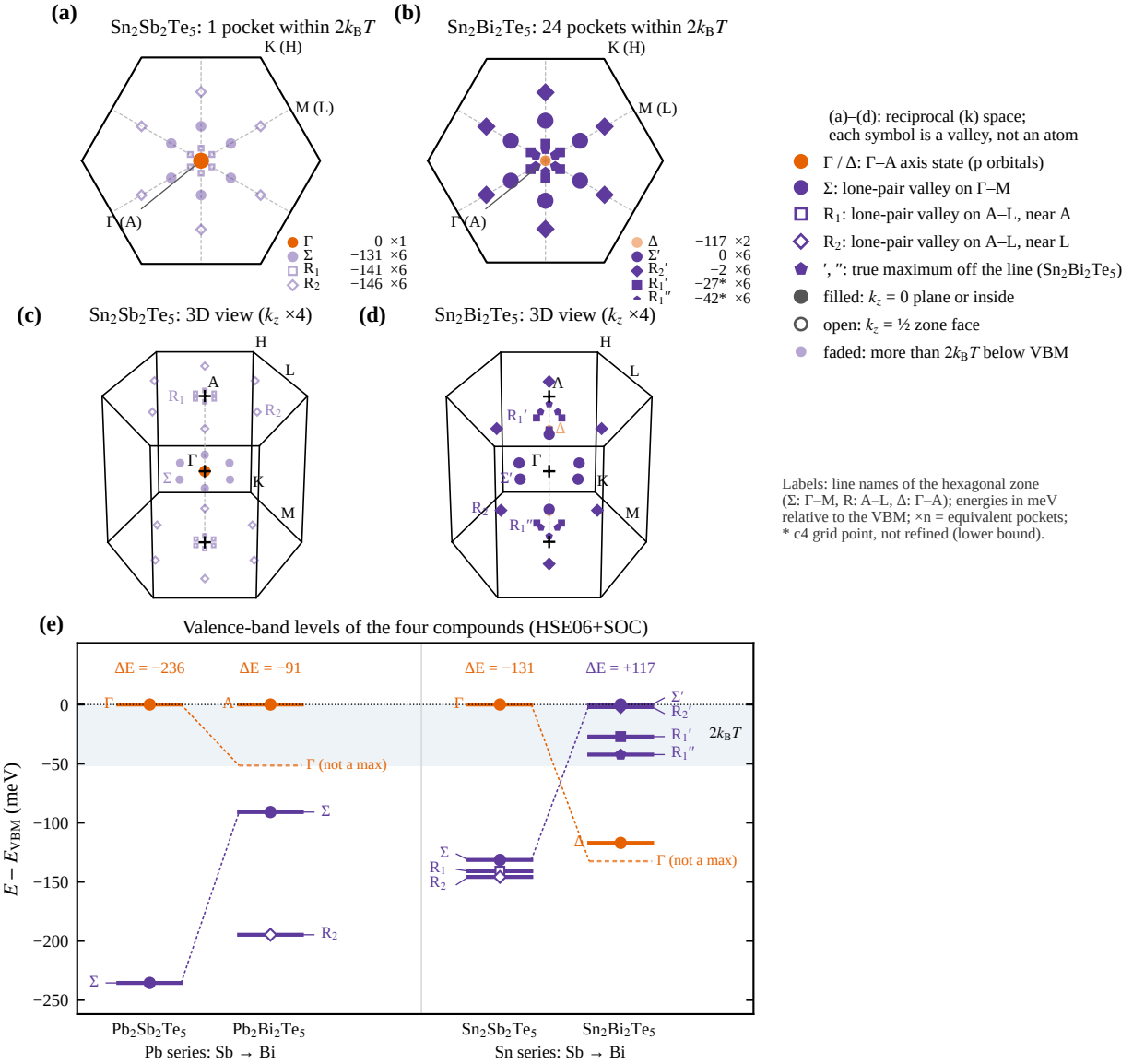
(a)
Sn2Sb2Te5: 1 pocket within 2kBT
K (H)
M (L)
Γ (A)
Γ 0 ×1
Σ −131 ×6
R1 −141 ×6
R2 −146 ×6
(b)
Sn2Bi2Te5: 24 pockets within 2kBT
K (H)
M (L)
Γ (A)
Δ −117 ×2
Σ′ 0 ×6
R2′ −2 ×6
R1′ −27* ×6
R1″ −42* ×6
(a)–(d): reciprocal (k) space;
each symbol is a valley, not an atom
Γ / Δ: Γ–A axis state (p orbitals)
Σ: lone-pair valley on Γ–M
R1: lone-pair valley on A–L, near A
R2: lone-pair valley on A–L, near L
′, ″: true maximum off the line (Sn2Bi2Te5)
filled: kz = 0 plane or inside
open: kz = ½ zone face
faded: more than 2kBT below VBM
(c)
Sn2Sb2Te5: 3D view (kz ×4)
H
L
A
R1
R2
Γ
Σ
K
M
(d)
Sn2Bi2Te5: 3D view (kz ×4)
H
L
A
R1′
Δ
Γ
Σ′
K
R2
R1″
M
Labels: line names of the hexagonal zone
(Σ: Γ–M, R: A–L, Δ: Γ–A); energies in meV
relative to the VBM; ×n = equivalent pockets;
* c4 grid point, not refined (lower bound).
(e)
Valence-band levels of the four compounds (HSE06+SOC)
ΔE = −236
ΔE = −91
ΔE = −131
ΔE = +117
Γ
A
Γ (not a max)
Σ
R2
2kBT
Σ′
R2′
R1′
R1″
Δ
Γ (not a max)
Σ
R1
R2
0
−50
−100
−150
−200
−250
E − EVBM (meV)
Pb2Sb2Te5
Pb2Bi2Te5
Pb series: Sb → Bi
Sn2Sb2Te5
Sn2Bi2Te5
Sn series: Sb → Bi